Year 11 · AQA Level 2 Further Maths · Covered topic

# Indices: fractional and negative; index equations

Revisit and practise · OnMaths · Created by Mr Ali

# Today I will learn to...

Goals

1. Evaluate fractional and negative indices

2. Solve equations with indices

# Do now: retrieval practice

4 mins

**1** Work out $2^5$

32

**2** Simplify $x^3 \times x^4$

$x^7$

**3** Simplify $(x^2)^3$

$x^6$

**4** Work out $\sqrt{81}$

9

Objective 1

# Evaluate fractional and negative indices

# Key idea

Learn

!

$a^{-n} = 1/a^n$. $a^{1/n}$ is the nth root of a. $a^{m/n}$: take the root first, then the power. $a^0 = 1$.

*Objective 1: Evaluate fractional and negative indices*

# I do: example 1

I do

**Work out $27^{2/3}$**

1. The cube root of 27 is 3
2. $3^2 = 9$
3. $27^{2/3} = 9$

# I do: example 2

I do

**Work out $16^{-3/4}$**

1. Negative power: $1/16^{3/4}$
2. The fourth root of 16 is 2
3. $2^3 = 8$
4. $16^{-3/4} = 1/8$

# I do: example 3

I do

**Work out $(8/125)^{-2/3}$**

1. Negative power: flip the fraction to $(125/8)^{2/3}$
2. Cube roots: 5/2
3. Square: 25/4
4. $(8/125)^{-2/3} = 25/4$

# Mini whiteboards

We do

1. Work out $32^{3/5}$

Answer: 8

2. Work out $(4/9)^{-1/2}$

Answer: 3/2

3. Work out $1000^{2/3}$

Answer: 100

4. Work out $9^{-3/2}$

Answer: 1/27

# You do: choose your challenge

You do

**Bronze**

Work out $49^{1/2}$

**Silver**

Work out $8^{-2/3}$

**Gold**

Work out $(27/64)^{2/3}$

**Platinum**

Simplify $(25x^4y^{-2})^{-1/2}$

# You do: answers

Check

| | |
|---|---|
| Bronze | 7 |
| Silver | 1/4 |
| Gold | 9/16 |
| Platinum | $y/(5x^2)$ |

# If you finish early

Extra

**Bronze**

Work out $125^{1/3}$

5

**Silver**

Work out $4^{-5/2}$

1/32

**Gold**

Work out $(81/16)^{-3/4}$

8/27

**Platinum**

Simplify $(8a^{9}b^{-6})^{1/3}$

$2a^{3}/b^{2}$

# Exam question 1

2 marks

Work out $64^{-2/3}$

*Space for working*

**Mark scheme: $1/64^{2/3}$ (M1) = 1/16 (A1)**

# Exam question 2

2 marks

Simplify $(16x^8)^{3/4}$

*Space for working*

**Mark scheme: $16^{3/4} = 8$ (M1). $8x^6$ (A1)**

Objective 1 achieved

# Evaluate fractional and negative indices



Objective 2

# Solve equations with indices

✓ Achieved: Evaluate fractional and negative indices

# Key idea

Learn

!

Write both sides as powers of the same base. Then the powers must be equal. For example $8 = 2^3$ and $1/4 = 2^{-2}$.

*Objective 2: Solve equations with indices*

# I do: example 1

I do

**Solve $2^x = 32$**

1. $32 = 2^5$
2. $2^x = 2^5$
3. $x = 5$

# I do: example 2

I do

**Solve $9^x = 27$**

1. $9 = 3^2$ and $27 = 3^3$
2. $3^{2x} = 3^3$
3. $2x = 3$
4. $x = 3/2$

# I do: example 3

I do

**Solve $4^{x+1} = 8^{x}$**

1. $4 = 2^2$ and $8 = 2^3$
2. $2^{2(x+1)} = 2^{3x}$
3. $2x + 2 = 3x$
4. $x = 2$

# Mini whiteboards

We do

1. Solve $3^x = 1/81$

Answer: $x = -4$

2. Solve $25^x = 125$

Answer: $x = 3/2$

3. Solve $2^x = 1/16$

Answer: $x = -4$

4. Solve $8^x = 2$

Answer: $x = 1/3$

# You do: choose your challenge

You do

**Bronze**

Solve $5^x = 125$

**Silver**

Solve $2^x = 1/8$

**Gold**

Solve $4^x = 32$

**Platinum**

Solve $9^{x-1} = 27^x$

# You do: answers

Check

**Bronze**

$x = 3$

**Silver**

$x = -3$

**Gold**

$2^{2x} = 2^5$, so $x = 5/2$

**Platinum**

$3^{2x-2} = 3^{3x}$, so $x = -2$

# If you finish early

Extra

**Bronze**

Solve $7^x = 49$

$x = 2$

**Silver**

Solve $10^x = 0.001$

$x = -3$

**Gold**

Solve $27^x = 3^{x+4}$

$3x = x + 4,\ x = 2$

**Platinum**

Solve $2^{3x-1} = 4^{x+1}$

$3x - 1 = 2x + 2,\ x = 3$

# Exam question 1

2 marks

Solve $2^x \times 4 = 64$

*Space for working*

**Mark scheme: $2^{x+2} = 2^6$ (M1). x = 4 (A1)**

# Exam question 2

3 marks

Solve $27^{x} = 9^{x+1}$

*Space for working*

**Mark scheme: $3^{3x} = 3^{2(x+1)}$ (M1). $3x = 2x + 2$ (M1). $x = 2$ (A1)**

Objective 2 achieved

# Solve equations with indices



# Real life

Apply

Bacteria double every hour: $N = 100 \times 2^t$. After how many hours will there be 6 400 bacteria?

**$2^t = 64 = 2^6$, so t = 6 hours**

# Extension: stretch yourself

Stretch

1. Solve $4^x - 2^x - 12 = 0$. (Hint: let $y = 2^x$.)

$y^2 - y - 12 = 0$, $(y - 4)(y + 3) = 0$. $2^x = 4$ so $x = 2$ ($2^x = -3$ is impossible)

2. Write $8^x \times 2^y$ as a single power of 2

$2^{3x + y}$

# Exit ticket: exam questions

Exit

1. Work out $81^{3/4}$ **(1 mark)**

2. Work out $125^{-2/3}$ **(2 marks)**

3. Solve $8^{x} = 4^{x+2}$ **(2 marks)**

# Exit ticket: mark scheme

Mark it

**Q1** 27 [1]

**Q2** $1/125^{2/3}$ (M1) = 1/25 (A1) [2]

**Q3** $2^{3x} = 2^{2x+4}$ (M1). x = 4 (A1) [2]

**Total: _____ / 5**

## Today I achieved

- ✔ **Evaluate fractional and negative indices**
- ✔ **Solve equations with indices**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

# Homework: exam questions

Due next lesson

1. Work out $36^{-1/2}$ [1]

2. Work out $(16/81)^{3/4}$ [2]

3. Simplify $(8x^6)^{2/3}$ [2]

4. Solve $3^{x+1} = 243$ [2]

5. Solve $16^x = 8^{x+1}$ [3]

*Extra practice: Mr Barton Maths: AQA Further Maths → search "Fractional indices"*